Friction Is Directly Proportional to the Normal Force

SYNOPSIS

Without friction, you can't walk. To move forward you push *backward* on the ground, and it is the ground's grip — friction — that pushes you forward in return: Newton's third law in action. That grip depends on just two things: the normal force pressing the surfaces together, and the surface roughness, μ. Note carefully which one: the *normal force*, not the weight. On a level floor they happen to be equal, which is why the two are so easily confused. And note the odd geometry — friction and the normal force are directly proportional yet *perpendicular* to each other: the one case where a vertical force sets the size of a horizontal one.

BEFORE YOU READ ON — PREDICT

You drag a loaded sled across the bench at a steady speed, pulling it with a spring scale, and the scale reads some value. Now you double the mass riding on the sled and drag it again, the same way.

What does the scale read — the same, twice as much, or somewhere in between? And what happens to the *coefficient* of friction?

Commit to an answer for both before you turn the page. You will measure it yourself.

PERFORMANCE EXPECTATIONS

1. **State** the direction of the friction force for a given relative motion, including the case of walking.
2. **Compute** the friction force from μ and the normal force, and the normal force on a level surface.
3. **Explain** what μ does and does not depend on — and why the normal force, not the weight, is what friction follows.
4. **Decide** whether static friction is at its maximum or below it in a described situation.
5. **Measure** a coefficient of kinetic friction as the slope of a friction-vs-normal-force graph.

6. **Use** the static-to-kinetic force trace to say what happens at the instant sliding begins.

Your tasks in this note

1. **Free-body diagram:** a person walking forward — where does friction point, and why is that the direction of motion?
2. **Free-body diagram:** a falling parachuter — drag is friction's cousin acting through a fluid.
3. **Write the equations** of kinetic and static friction in terms of μ_k, μ_s, and the normal force.
4. **Work the questions** in your guided notes as the video develops each case.

The key relationships

Eq. 1	$f_k = \mu_k N$	kinetic friction — surfaces sliding past each other
Eq. 2	$f_s \leq \mu_s N$	static friction — grips as hard as needed, up to its limit
Eq. 3	$\mu_s > \mu_k$	breaking the grip is harder than keeping the slide — why wheels lock and joints creak loose
Eq. 4	$f = \mu m g$	**level surface only** — the shortcut you get by substituting N = mg

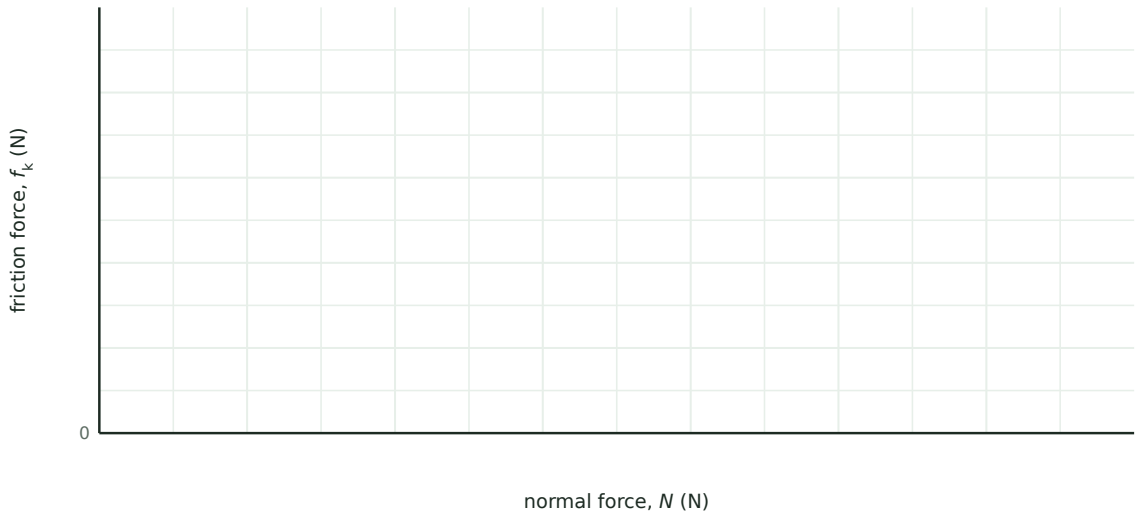
Measure μ for yourself

A coefficient of friction is not something to look up — it is something you can measure in about ten minutes. You will drag a loaded sled across the bench with a spring scale and find μ_k as the slope of a graph.

1. **Set up.** Put the sled on the bench with the spring scale hooked to it, the scale held horizontally.
2. **Pull at a steady speed.** This part matters more than anything else in the procedure. Only when the sled moves at *constant velocity* is the acceleration zero, so that the forces balance and the scale reading equals the friction force exactly. If you are speeding it up, the reading is bigger than the friction and your data is wrong. Practise until you can hold it steady.
3. **Watch the scale as it breaks loose.** Pull slowly from rest and note the reading at the instant the sled starts to move — then the lower, steadier reading once it is sliding. Record both.

4. **Add mass and repeat.** Load the sled with five different masses, dragging steadily each time, and record the scale reading for each.

LOAD	TOTAL MASS, M (KG)	NORMAL FORCE, N = MG (N)	SCALE READING WHILE SLIDING = F_K (N)
sled alone			
+ mass 1			
+ mass 2			
+ mass 3			
+ mass 4			



Plot the friction force against the normal force. On a level bench the normal force is just the total weight, N = mg — so the third column of your table is the x-axis and the fourth is the y-axis.

1. **Is your plot a straight line through the origin?** If it is, you have just shown that friction is *directly proportional* to the normal force — which is the title of this note, and now it is your measurement rather than my claim.

2. **Find the slope.** What are its units? The slope is μ_k, and the answer to the units question is why μ is just a number.

3. **Compare your two readings from step 3** — the one at the instant it broke loose, and the steady one while sliding. Which is bigger, and what does that tell you about μ_s and μ_k?

4. **Now turn the sled on its side**, so it rests on a smaller face, and drag the same load again. Predict first, then measure. Does the reading change?

5. **Now check it.** You doubled the mass. What happened to the scale reading, and what happened to μ?

BACK TO YOUR PREDICTION

You predicted what the spring scale would read when you doubled the mass on the sled. Compare that with what your data actually showed, and name the quantity you were really changing.

WATCH · THE NOTE WORKED THROUGH

Recording to come. Print the guided notes (link just below), watch, and fill them in as you go.

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